

The Fill Point

Architecture, Evidence, and the YouTube Primitive Pool

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Abstract

Viral “primitive construction” videos frequently culminate in the filling of a carved earthen pool. This essay argues that the fill is not merely a visual climax. It is a technical threshold that changes what the object must be able to do and therefore what evidence becomes relevant. Before water, a finished basin can be judged primarily as geometry: a void has been excavated, trimmed, and surfaced. After water, the same object makes claims about material composition, erosion resistance, cracking, retention, drainage, and duration. Research on rock-cut architecture shows that subtractive building depends on the material properties of the excavated context, while research on earthen construction shows that water exposure and long-term weathering are decisive to durability. Read against these literatures, the genre’s recurrent narrative endpoint becomes architecturally peculiar: the videos often stop at the moment when a building would begin accumulating its strongest performance evidence. I call this the *fill point*. The concept shifts criticism away from the exhausted binary of “real” versus “fake” and toward a forensic architectural question: which operations are visible because they produce the image of completion, and which technical systems remain unresolved because their evidence would only appear after completion?

Keywords: YouTube; primitive construction; earthen architecture; excavation; pools; durability; platform architecture; construction media

This manuscript was assembled with AI assistance from a preserved research corpus. A complete provenance record, source map, and assembly prompt accompany the publication.

1 Introduction

The most striking objects in the viral “primitive construction” genre are difficult to describe with the vocabulary the genre supplies. They are presented as primitive shelters or survival constructions, yet their recurrent programs are not restricted to shelter. They include sunken living rooms, swimming pools, moats, water slides, bridges, sculpted furniture, terraces, grottos, and underground villas. The characteristic architectural operation is similarly unusual. Rather than beginning with a frame and adding enclosure, many projects begin with a continuous mass of ground and remove material until rooms, stairs, ledges, basins, and platforms remain.

This has encouraged two common readings. One treats the videos as demonstrations of primitive ingenuity; the other treats them as frauds whose hidden machinery, labor, cement, or staging invalidates the image. Both readings make authenticity the decisive question. The first asks whether the displayed work is genuinely primitive; the second asks whether the stated means match the actual means. Those questions matter, but they do not exhaust the architectural problem.

A building can be materially real while its implied construction system is misleading. Hand trimming can be genuine even if bulk excavation occurred off camera. A carved wall can be earth while its water-contact surface is stabilized or rendered. A pool can hold water for a filmed interval without establishing how it handles erosion, leakage, rainfall, or maintenance. The crucial distinction is therefore not simply between real and fake. It is between *visible completion* and *technical continuation*.

This essay develops that distinction through a single recurring event: filling the pool. Before the fill, the object can remain ambiguous between sculpture, excavation, and architecture. After the fill, water imposes new technical demands and makes previously invisible variables consequential. Paradoxically, this is often where the video reaches closure. The fill is simultaneously a narrative ending and an architectural beginning.

2 Subtractive Form Is a Material Claim

Rock-cut architecture provides a useful comparison because it makes subtraction explicit as a building technique. Mangeli et al. define rock-cut building as the creation of space by hollowing a natural solid context rather than assembling conventional building materials. Crucially, they do not treat any excavatable ground as equivalent. The context must be workable with available tools while remaining sufficiently coherent not to fall apart, and excavation quality depends in part on the material of the settlement context and its particle characteristics.¹

That distinction complicates the easy description of YouTube pools as “carved earth.” Excavation explains how a void is produced; it does not by itself explain why the mass that remains can function as a wall, stair, shelf, roof, retaining surface, or water-bearing shell. In rock-cut architecture, structural continuity is inherited from rock. In an excavated soil structure, continuity must be investigated rather than assumed. Cohesion, density, moisture, layering, cut height, geometry, stabilization, and duration all become potentially load-bearing variables.

The point is not that earthen architecture is inherently unstable. It is not. Durable earthen

¹Mohammad Mangeli et al., “A New Look at Excavation Techniques and Design of Rock-Cut Architectures,” *Designs* 6, no. 4 (2022): 64, <https://doi.org/10.3390/designs6040064>.

buildings exist across many climates and techniques. Bui et al. document stabilized and unstabilized rammed-earth walls that remained standing after two decades of natural weathering.² The point is more exact: “earth” is not a sufficient material specification. A structure’s apparent monolith may conceal decisive differences in compaction, grading, additives, renders, membranes, or foundations.

This suggests a forensic method. Finished excavated geometry can be read backward. Mangeli et al. reconstruct historical excavation procedures at Meymand partly from surviving architectural fabric because documentary evidence of the process is scarce. The same logic can be applied cautiously to video architecture. Deep pits imply spoil-removal paths. Central islands imply either preserved mass or later deposition. Enclosed chambers impose access and sequencing constraints. Nested pools require ordered excavation and finishing. A video can omit an operation, but a finished geometry may still retain dependencies that the omitted operation had to satisfy.

3 The Fill Point

The most analytically useful moment arrives when water enters the basin. Narloch and Woyciechowski assess cement-stabilized rammed earth through tests that include wet-to-dry strength, shrinkage, pressurized-water erosion, and cyclic wetting and drying.³ Their specific system differs from an excavated YouTube pool, but the variables matter because water changes the question one can ask of an earthen surface.

Before filling, the camera can establish a shape: a basin has a perimeter, a depth, stairs, ledges, perhaps a slide. Surface smoothness can be seen but not yet operationally tested. Once water is introduced, the object becomes a water-bearing assembly. Permeability, erosion resistance, cracking, retention, overflow, drainage, and maintenance become consequential whether or not they remain visible in the edit.

I call this transition the *fill point*. It is the handoff from sculptural geometry to infrastructure. The phrase does not imply that a conventional swimming-pool system must exist, nor that every basin is intended to be permanent. It names the moment at which the object’s evidentiary burden changes. A cavity can be visually complete while its hydraulic behavior remains unknown.

This is why the recurrent final swim is architecturally more significant than it first appears. The body entering the pool demonstrates scale, depth, and immediate usability. It can show that the slide conveys a person into the water and that the basin holds water long enough to film. But commissioning is not durability. The final swim supplies strong evidence for a narrow interval and almost none for the sequence that follows: rainfall, saturation, drying, erosion, cracking, biological growth, maintenance, repair, or abandonment.

4 The Architecture After the Ending

Bui et al.’s twenty-year study makes a simple but consequential point: some architectural evidence is delayed. Weathering produces information that a completion photograph cannot contain. Their study measured material loss after long exposure and found different erosion behavior in stabilized

²Q. B. Bui et al., “Durability of rammed earth walls exposed for 20 years to natural weathering,” *Building and Environment* 44, no. 5 (2009): 912–919, <https://doi.org/10.1016/j.buildenv.2008.07.001>.

³Piotr Narloch and Piotr Woyciechowski, “Assessing Cement Stabilized Rammed Earth Durability in A Humid Continental Climate,” *Buildings* 10, no. 2 (2020): 26, <https://doi.org/10.3390/buildings10020026>.

and unstabilized walls. The important lesson for the present problem is methodological rather than directly comparative. A building's performance is a time series.

The primitive-pool video often adopts a different temporal logic. Construction is narrated as a sequence of visible transformations culminating in a finished state. Digging, shaping, smoothing, filling, and occupation move toward closure. Once the object reads as complete, the video has achieved its narrative task. Architecture, however, does not stop accumulating consequences when it becomes photographically complete.

This difference allows a more precise critique than “the structures are fake.” A short-lived set can be materially real. A temporary basin can satisfy its intended use. A stabilized earthen surface can be legitimate construction. What remains unresolved is the duration and performance claim implied by the representation. If the video invites the viewer to infer a durable inhabitable pool while supplying evidence only through the first minutes after filling, the problem lies in the mismatch between temporal evidence and architectural inference.

That mismatch can be tested. The strongest future study would not begin by searching for excavators. It would build an afterlife archive. For identifiable sites, researchers could compare the final reveal with later imagery after rain, a wet season, or a year. Photogrammetric comparison could quantify erosion. Water levels could reveal leakage or drainage. Surface cracking, repairs, vegetation, collapse, reuse, or abandonment would each become evidence. The architecture after the ending is not a footnote to the video; it is where competing explanations become distinguishable.

5 Material Affordance Versus Platform Selection

The recurrence of circles, spirals, nested pools, dramatic section changes, and overhead-readable compositions raises a second problem: why these forms? It is tempting to attribute them directly to YouTube. Grupa and Nalaskowski's study of the broader “YouTube cavemen” phenomenon identifies rapid proliferation of related channels and argues that viewer attractiveness increasingly affects content selection.⁴ Hallinan likewise shows how creators can optimize cultural production around perceived platform values and formal presentational practices.⁵

But platform selection should not be used as a total explanation. Excavation itself has affordances. A curve can be generated by removing material rather than fabricating many differently angled components. Sectional descent can arise from the construction process before it becomes an effective reveal. High-angle views may exploit an architecture whose spatial logic was already produced by digging. Media form and material sequence may reinforce one another without either being the sole origin.

This creates a tractable empirical question. A chronological dataset could code plan geometry, depth, water program, camera angle, thumbnail composition, normalized view performance, and subsequent projects. If unusually successful traits recur disproportionately in later builds, platform selection becomes more plausible. If the same traits track soil conditions or excavation technique regardless of view performance, material affordance gains explanatory weight. The distinction matters because

⁴Małgorzata Grupa and Filip Nalaskowski, “‘YouTube Cavemen.’ The phenomenon of channels presenting technological development in ‘prehistory’,” *Raport* 14 (2019): 129–135.

⁵Blake Hallinan, “No judgment: value optimization and the reinvention of reviewing on YouTube,” *Journal of Computer-Mediated Communication* 28, no. 5 (2023): zmad034, <https://doi.org/10.1093/jcmc/zmad034>.

resemblance is not mechanism.

6 A Cambodian Location Is Not Yet a Cambodian Genealogy

The architectural genealogy also remains unsettled. The presence of many channels in Southeast Asia, including Cambodia, does not make their pits and pools examples of local vernacular architecture by default. The Center for Khmer Studies' *Wooden Architecture of Cambodia* documents wood building as a major, historically significant Cambodian tradition and emphasizes its techniques together with social and symbolic dimensions.⁶

The viral pool morphology is strikingly different: mass rather than frame, descent rather than elevation, grotto rather than lightweight enclosure, sculpted bench rather than assembled furniture, pool and water slide rather than the climatic and domestic logic of the timber house. That contrast does not establish where the online forms came from. It establishes only that location is not lineage.

A proper genealogy would therefore compare individual transferable elements rather than broad visual similarity. Roof profiles, portals, stair types, pool geometries, symmetry, ornament, furniture, circulation, and sectional organization could be tested against documented Khmer domestic architecture, Angkorian monumental vocabularies, contemporary resorts and water parks, games, miniature-building media, and earlier channels in the same recommendation ecology. Until such work is done, “primitive Cambodian architecture” is too blunt a category.

7 Discussion: From Authenticity to Evidentiary Architecture

The fill point reorganizes the field because it joins three questions that are usually kept separate.

First is construction: what operations and materials make the visible geometry possible? Second is media: which stages are selected as legible transformation and closure? Third is duration: what evidence exists only after the video has finished? The pool is where these questions collide. Water is both a rendering material—it produces contrast, reflection, movement, sound, and an immediate sense of completion—and a material test that increases the demands placed on the substrate and surface.

This dual role explains why the pool is a better research object than the generic “primitive hut.” A hut can remain difficult to diagnose from appearance alone. A water-bearing cavity imposes a sharper threshold. Either the object has a system adequate to its intended duration or it does not. The system may be raw cohesive earth, stabilized soil, render, membrane, concrete, repeated maintenance, or simply the short time horizon of the shoot. The crucial task is to identify which.

The argument also alters the meaning of hidden machinery. An excavator, if present, is one kind of omitted operation, but it is not necessarily the most architecturally important one. The decisive hidden layer may be the surface treatment, the soil choice, the drainage path, the curing interval, or the temporal limit. “How was the hole dug?” is only one question. “What makes the filled object continue to be what the video says it is?” is harder.

⁶François Tainturier, ed., *Wooden Architecture of Cambodia: A Disappearing Heritage* (Phnom Penh: Center for Khmer Studies, 2006), <https://library.khmerstudies.org/bib/4921>.

8 Limitations

The present argument is a research program rather than a completed forensic study of a fixed video sample. It draws on an existing corpus of observed genre features and on scholarship in rock-cut and earthen architecture, Cambodian wooden architecture, and YouTube cultural production. It does not establish the material composition of any particular pool, the presence of machinery at a particular site, or the intended lifespan of a particular structure.

The comparison with rammed earth is deliberately limited. Rammed earth is compacted in formwork and is not equivalent to an excavated soil wall. Its relevance lies in the variables that durability research makes explicit, especially water exposure, stabilization, erosion, and time. Likewise, research on rock-cut architecture concerns geological contexts that cannot simply be transferred to tropical soils. It shows why substrate matters, not what a Cambodian substrate necessarily is.

Finally, the claim that platform incentives affect architectural morphology remains hypothetical. Existing scholarship establishes creator optimization and genre proliferation, not a causal chain from recommendation systems to plan geometry. That chain requires chronological, comparative data.

9 Conclusion

The most productive question to ask of YouTube primitive construction is not whether the finished pool is real. It is what kind of evidence would make the pool intelligible as architecture.

The genre’s signature operation is subtraction: ground becomes void, void becomes form, form becomes smooth surface, and surface becomes pool. Yet each transition introduces variables that the visible sequence can underdescribe. Subtraction requires a material capable of remaining. Surface continuity may conceal a composite system. Water converts geometry into a hydraulic claim. Weather and use generate evidence after the reveal.

The fill point names the genre’s central temporal inversion. What the video treats as completion is the moment when architecture begins its strongest cross-examination. That inversion offers a concrete research method: reconstruct predecessor operations from geometry, characterize substrate and surface, document the fill, and then keep watching after the video stops.

References

- Bui, Q. B., J.-C. Morel, B. V. Venkatarama Reddy, and W. Ghayad. 2009. “Durability of rammed earth walls exposed for 20 years to natural weathering.” *Building and Environment* 44(5): 912–919. <https://doi.org/10.1016/j.buildenv.2008.07.001>.
- Grupa, Małgorzata, and Filip Nalaskowski. 2019. “‘YouTube Cavemen.’ The phenomenon of channels presenting technological development in ‘prehistory.’” *Raport* 14: 129–135.
- Hallinan, Blake. 2023. “No judgment: value optimization and the reinvention of reviewing on YouTube.” *Journal of Computer-Mediated Communication* 28(5): zmad034. <https://doi.org/10.1093/jcmc/zmad034>.
- Mangeli, Mohammad, Farshid Aram, Reza Abouei, and Fatemeh Mehdizadeh Saradj. 2022. “A New Look at Excavation Techniques and Design of Rock-Cut Architectures.” *Designs* 6(4):64.

<https://doi.org/10.3390/designs6040064>.

Narloch, Piotr, and Piotr Woyciechowski. 2020. "Assessing Cement Stabilized Rammed Earth Durability in A Humid Continental Climate." *Buildings* 10(2):26. <https://doi.org/10.3390/buildings10020026>.

Tainturier, François, ed. 2006. *Wooden Architecture of Cambodia: A Disappearing Heritage*. Phnom Penh: Center for Khmer Studies. <https://library.khmerstudies.org/bib/4921>.

Assembly Appendix

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Exact assembly prompt filename: `SLIPCASE_FINAL_PROMPT.txt`

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